Machine Learning and Deep Learning Lecture-01 [Exercise]

Spring 2026

Implementation Steps

- Step One:

Load Required Packages

- Step Two:
  - CSV
  - URL
  - Package

Load Required Data

Implementation Steps

- Step Three:

Write your Code:
Build model

- Step Four:
  - Table
  - Plots
  - Histogram
  - Bar Diagram etc.

Present your Result

How to learn APIs?

```
>>> import numpy as np
>>> X = np.array([[-1, -1], [-2, -1], [-3, -2], [1, 1], [2, 1], [3, 2]])
>>> Y = np.array([1, 1, 1, 2, 2, 2])
>>> from sklearn.naive_bayes import GaussianNB
>>> clf = GaussianNB()
>>> clf.fit(X, Y)
GaussianNB()
>>> print(clf.predict([[-0.8, -1]]))
[1]
>>> clf_pf = GaussianNB()
>>> clf_pf.partial_fit(X, Y, np.unique(Y))
GaussianNB()
>>> print(clf_pf.predict([[-0.8, -1]]))
[1]
```

https://scikit-learn.org/stable/tutorial/index.html

sklearn.naive_bayes.GaussianNB

class `sklearn.naive_bayes.GaussianNB`(*, *priors=None*, *var_smoothing=1e-09*) [source]

Gaussian Naive Bayes (GaussianNB).

Can perform online updates to model parameters via `partial_fit`. For details on algorithm used to update feature means and variance online, see Stanford CS tech report STAN-CS-79-773 by Chan, Golub, and LeVeque:

http://i.stanford.edu/pub/cstr/reports/cs/tr/79/773/CS-TR-79-773.pdf

Read more in the User Guide.

Parameters:

priors : ***array-like of shape (n_classes,), default=None***
Prior probabilities of the classes. If specified, the priors are not adjusted according to the data.

var_smoothing : ***float, default=1e-9***
Portion of the largest variance of all features that is added to variances for calculation stability.

New in version 0.20.

Attributes:

class_count_ : ***ndarray of shape (n_classes,)***
number of training samples observed in each class.

class_prior_ : ***ndarray of shape (n_classes,)***
probability of each class.

classes_ : ***ndarray of shape (n_classes,)***
class labels known to the classifier.

epsilon_ : ***float***
absolute additive value to variances.

n_features_in_ : ***int***
Number of features seen during fit.

New in version 0.24.

feature_names_in_ : ***ndarray of shape (`n_features_in_`,)***
Names of features seen during fit. Defined only when `X` has feature names that are all strings.

New in version 1.0.

var_ : ***ndarray of shape (n_classes, n_features)***
Variance of each feature per class.

New in version 1.0.

theta_ : ***ndarray of shape (n_classes, n_features)***
mean of each feature per class.

Hands-on Session

Plan - A

- Post Exercise in Canvas.
- Explain the exercise.
- Show Part of code in 10 mins (before 1st break)
- Show remaining code in 10 mins (before end of the session).

Plan - B

- Post Exercise in Canvas.
- Explain the exercise.
- Live Coding before end of the session.

TAs are there to help.